



WAZUH SIEM DEPLOYMENT AND CONFIGURATION

Centralized Security Monitoring using Wazuh SIEM

File Integrity Monitoring (FIM) Implementation and Security Event Validation

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Project Title

Deployment and Configuration of Wazuh SIEM with Windows and Linux Endpoints

Table of Contents

1. Objective
2. Introduction
3. Lab Environment
4. Network Topology
5. Implementation Overview
 - 5.1 Deploy Wazuh Virtual Appliance
 - 5.2 Configure Wazuh Manager
 - 5.3 Install Wazuh Agent on Windows
 - 5.4 Register Windows Agent
 - 5.5 Install Wazuh Agent on Linux
 - 5.6 Register Linux Agent
 - 5.7 Configure File Integrity Monitoring (FIM)
 - 5.8 Generate Security Events
 - 5.9 Verify Security Alerts
6. Challenges Faced
7. Learning Outcomes
8. Conclusion
9. References
10. Appendix

1. Objective

The objective of this project is to deploy and configure a centralized Security Information and Event Management (SIEM) environment using Wazuh.

The project demonstrates how a Security Operations Center (SOC) monitors multiple endpoints from a single management server by collecting, analyzing, and correlating security events.

The implementation involves deploying the Wazuh Virtual Appliance, configuring the Wazuh Manager, installing Wazuh Agents on both Windows and Linux endpoints, registering the agents with the manager, configuring File Integrity Monitoring (FIM), generating security events, and verifying that alerts are successfully detected within the Wazuh Dashboard.

This project provides practical experience in endpoint monitoring, centralized log management, event correlation, and security monitoring techniques that are commonly used in modern SOC environments.

2. Introduction

Cybersecurity organizations continuously monitor systems to detect unauthorized activities, malware infections, policy violations, and system modifications.

As organizations grow, manually reviewing logs from individual computers becomes inefficient.

A Security Information and Event Management (SIEM) solution addresses this challenge by collecting logs from multiple endpoints, analyzing them centrally, and generating alerts whenever suspicious activity is detected.

Wazuh is an open-source SIEM and Extended Detection and Response (XDR) platform designed to provide centralized security monitoring and endpoint protection.

It enables organizations to monitor Windows, Linux, macOS, cloud environments, containers, and network devices from a single dashboard.

Wazuh performs several security functions including log analysis, intrusion detection, vulnerability detection, security configuration assessment, malware detection, and File Integrity Monitoring (FIM).

In this project, a Wazuh Virtual Appliance was deployed as the central management server.

Two endpoints—a Windows 11 host and a Parrot Security Linux system—were configured with Wazuh Agents.

These agents continuously monitored their respective systems and forwarded security events to the Wazuh Manager for analysis.

File Integrity Monitoring was then configured to detect file creation, modification, and deletion events, demonstrating how a SOC analyst can identify unauthorized changes to monitored files.

The successful deployment of this lab provides a realistic understanding of how enterprise SOC environments collect, analyze, and respond to security events using centralized monitoring technologies.

3. Lab Environment

3.1 Hardware and Software Requirements

The Wazuh SIEM environment was deployed in a virtualized lab using Oracle VirtualBox.

A Wazuh Virtual Appliance was configured as the central monitoring server, while Windows 11 and Parrot Security OS acted as monitored endpoints.

Both endpoints were connected to the Wazuh Manager through the local network using Wazuh Agents.

The following hardware and software components were used during this project.

Component	Specification
Host Operating System	Windows 11 Pro
Virtualization Software	Oracle VirtualBox
Wazuh Server	Wazuh Virtual Appliance v4.14.6
Windows Endpoint	Windows 11 Pro
Linux Endpoint	Parrot Security OS 6.4
Web Browser	Mozilla Firefox
Network Mode	Bridged Adapter
Wazuh Manager IP Address	192.168.100.39
Windows Agent IP Address	192.168.100.34
Linux Agent IP Address	192.168.100.41

3.2 Components Used

The implemented environment consists of three major components.

- **Wazuh Manager:** The Wazuh Manager serves as the central server responsible for receiving logs from all connected agents. It analyzes security events, applies detection rules, generates alerts, and

stores event information for further investigation. The Wazuh Dashboard communicates with the manager to display alerts and security information to the analyst.

- **Wazuh Agent:** A Wazuh Agent is lightweight software installed on every monitored endpoint. The agent continuously collects system logs, monitors file changes, performs security checks, and securely forwards collected events to the Wazuh Manager for analysis.
- **Wazuh Dashboard:** The Wazuh Dashboard is a web-based interface used by SOC analysts to monitor the environment. It provides real-time visualization of security alerts, endpoint status, File Integrity Monitoring events, vulnerability information, and threat hunting capabilities.

4. Network Topology

The implemented network consists of one Wazuh Manager and two monitored endpoints. The Windows and Linux systems communicate with the Wazuh Manager through Wazuh Agents. All collected logs and security events are transmitted to the manager, where they are analyzed and displayed on the Wazuh Dashboard.

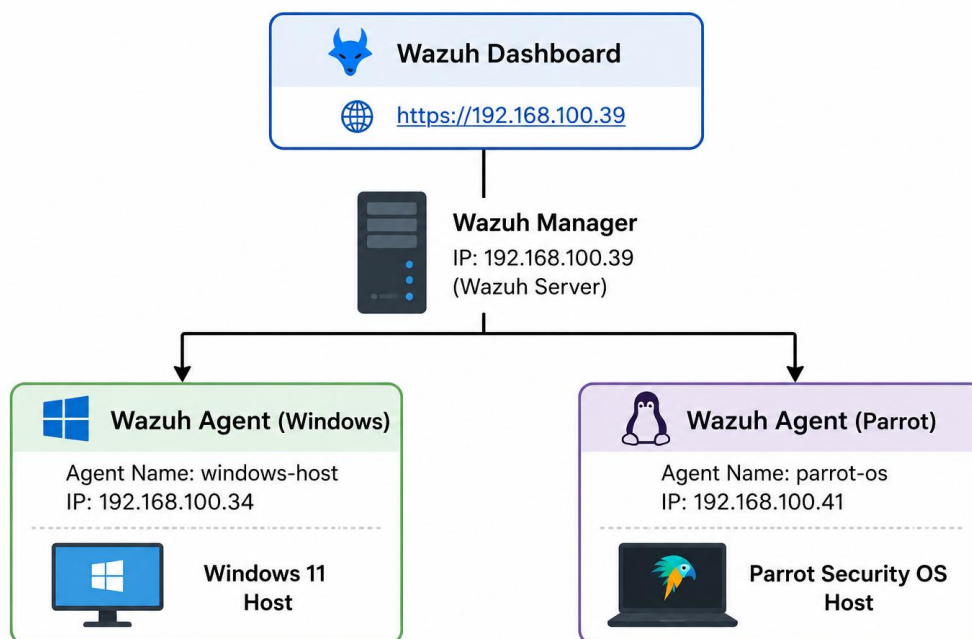


Figure 4.1: Network Topology of the Implemented Wazuh SIEM Environment.

4.1 Communication Workflow

The communication process within the implemented environment follows the sequence below:

- The Wazuh Agent monitors activities occurring on the endpoint.
- Security events such as log entries or file modifications are detected.
- The detected events are securely transmitted to the Wazuh Manager.
- The Wazuh Manager analyzes the received data using predefined rules.

- Alerts are generated whenever suspicious or monitored activities are identified.
- The Wazuh Dashboard displays the alerts, allowing SOC analysts to investigate the events.

This centralized architecture simplifies security monitoring and enables rapid detection of security incidents across multiple endpoints.

5. Implementation Overview

This chapter describes the complete deployment and configuration process carried out to build the Wazuh SIEM environment. The implementation was performed according to the project requirements provided by ITSimplera Institute.

The primary objective was to establish centralized monitoring by deploying a Wazuh Manager, connecting Windows and Linux endpoints, enabling File Integrity Monitoring (FIM), generating security events, and verifying that the alerts were successfully detected within the Wazuh Dashboard. The implementation was completed in the following sequence:

- Deployment of the Wazuh Virtual Appliance.
- Configuration of the Wazuh Manager.
- Installation of the Wazuh Agent on the Windows endpoint.
- Registration of the Windows endpoint with the Wazuh Manager.
- Installation of the Wazuh Agent on the Linux (Parrot Security OS) endpoint.
- Registration of the Linux endpoint with the Wazuh Manager.
- Configuration of File Integrity Monitoring (FIM).
- Generation of security events.
- Verification of alerts within the Wazuh Dashboard.

Each implementation step is documented with configuration details, commands executed, screenshots, and verification results. This approach ensures that the deployment process is reproducible and demonstrates a practical understanding of centralized security monitoring using Wazuh SIEM.

5.1 Deployment of the Wazuh Virtual Appliance

Objective

The first step of the implementation was to deploy the Wazuh Virtual Appliance, which serves as the central Security Information and Event Management (SIEM) server. The appliance contains the Wazuh Manager, Indexer, and Dashboard preconfigured, providing a complete platform for centralized security monitoring and log management.

Procedure

The Wazuh Virtual Appliance was imported into Oracle VirtualBox and configured with the recommended virtual hardware resources. The virtual machine was assigned a Bridged Network Adapter so that it could communicate directly with other systems on the local network.

After powering on the virtual machine, the operating system initialized all Wazuh services automatically. The assigned IP address of the Wazuh server was identified and used to access the web-based Wazuh Dashboard from the host computer. The dashboard was successfully accessed through a web browser using the server IP address.

Server Address: <https://192.168.100.39>

Result

The Wazuh Virtual Appliance was successfully deployed and became operational as the centralized SIEM server. The Wazuh Dashboard was accessible through a web browser, confirming that the Manager, Indexer, and Dashboard services were functioning correctly

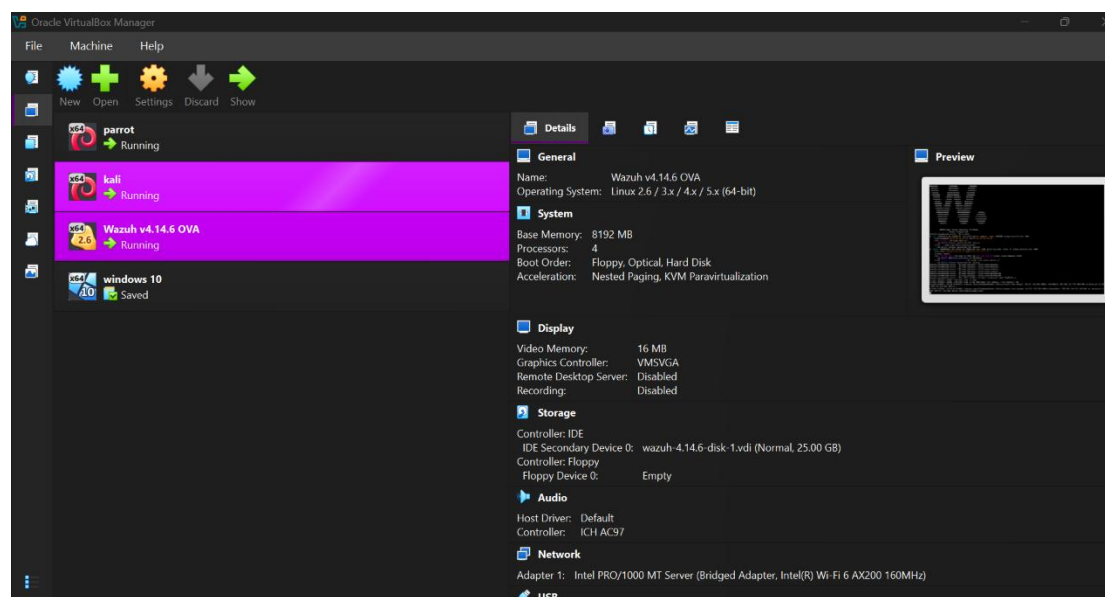


Figure 5.1: Wazuh Virtual Appliance running in Oracle VirtualBox.

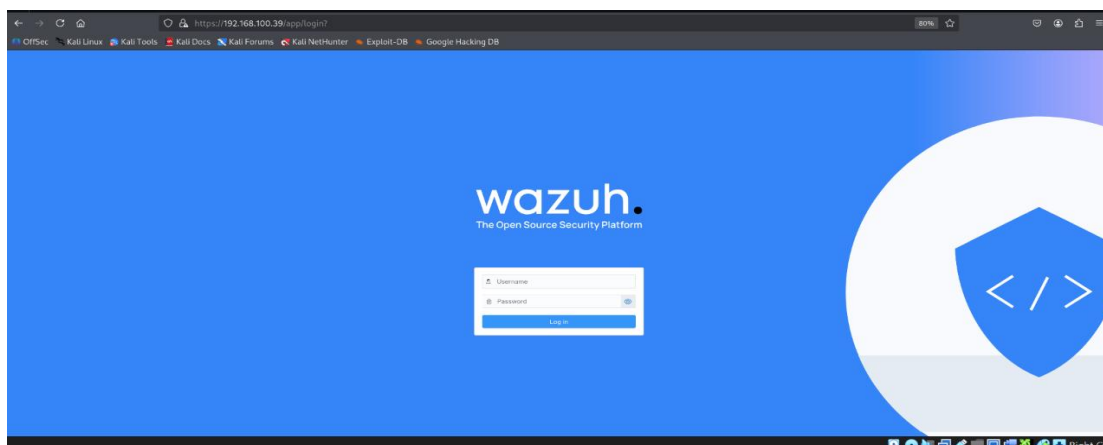


Figure 5.2:Wazuh Dashboard login

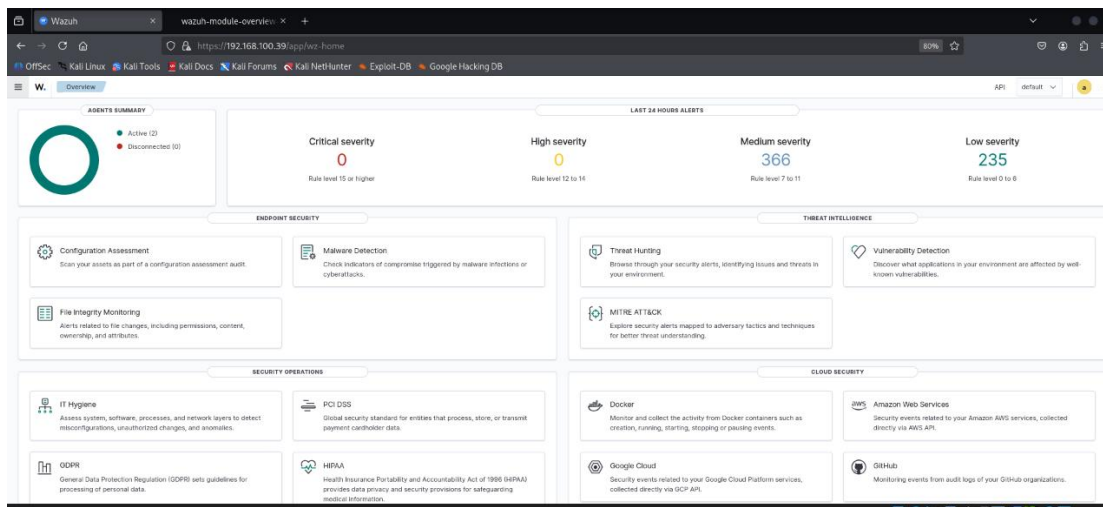


Figure 5.3:Wazuh Dashboard homepage

5.2 Configuration of the Wazuh Manager

Objective

After deploying the Wazuh Virtual Appliance, the Wazuh Manager was configured to receive security events from monitored endpoints. The manager acts as the central component responsible for collecting logs, analyzing events, applying detection rules, and generating security alerts.

Procedure

The Wazuh Dashboard was opened through the web interface, and the Manager services were verified to ensure they were running correctly. Network connectivity between the manager and the endpoints was confirmed using ICMP (ping).

The Wazuh Manager IP address used throughout the implementation was: **192.168.100.39**

This IP address was later configured in both Windows and Linux Wazuh Agents, allowing them to establish communication with the manager. The manager was verified to be reachable before proceeding with agent installation.

Result

The Wazuh Manager was successfully configured and ready to accept agent registrations and receive endpoint security events.

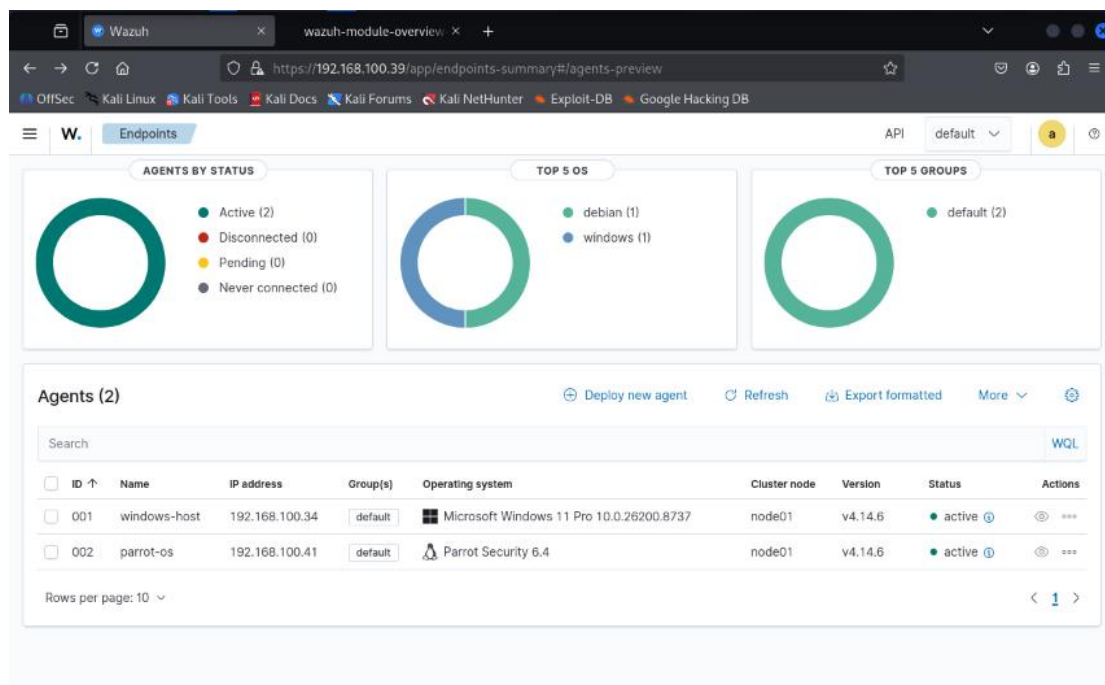


Figure 5.4: Wazuh Dashboard showing the Wazuh Manager in an operational state with connected endpoints.

5.3 Installation of the Wazuh Agent on Windows

The Wazuh Agent package was downloaded and installed on the Windows 11 host. The installation command was executed via PowerShell, specifying the manager IP address and assigning a descriptive host name to the agent.

5.4 Registration of the Windows Agent

Objective

The Windows endpoint was registered with the Wazuh Manager so that it could securely communicate with the central server and begin transmitting monitoring data.

Procedure

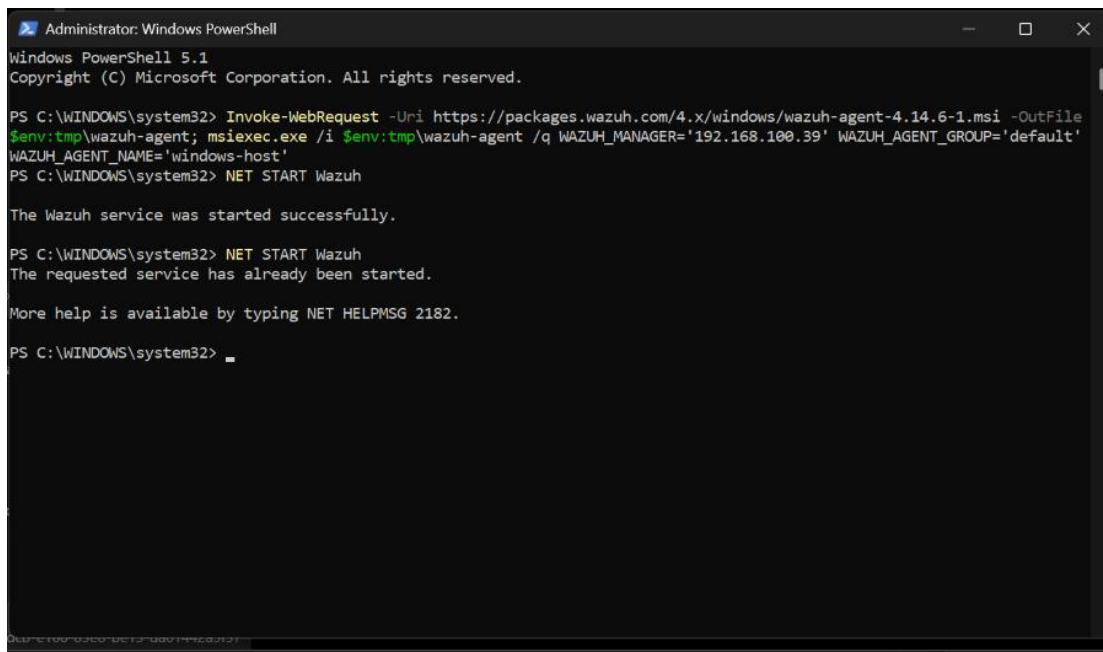
After starting the Wazuh Agent service, the endpoint automatically registered with the Wazuh Manager using the configured manager IP address. The Wazuh Dashboard was refreshed to verify successful registration. The registered endpoint appeared in the Endpoints section with the following information:

Property	Value
Agent Name	windows-host
Operating System	Windows 11 Pro
Status	Active
Manager	node01

The successful registration confirmed that communication between the Windows endpoint and the Wazuh Manager had been established.

Result

The Windows endpoint was successfully registered with the Wazuh Manager and appeared as Active in the Wazuh Dashboard.



```
Administrator: Windows PowerShell
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.6-1.msi -OutFile
$env:tmp\wazuh-agent; msexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.100.39' WAZUH_AGENT_GROUP='default'
WAZUH_AGENT_NAME='windows-host'
PS C:\WINDOWS\system32> NET START Wazuh

The Wazuh service was started successfully.

PS C:\WINDOWS\system32> NET START Wazuh
The requested service has already been started.

More help is available by typing NET HELPMSG 2182.

PS C:\WINDOWS\system32> _
```

Figure 5.5: Successful installation of the Wazuh Agent on Windows.

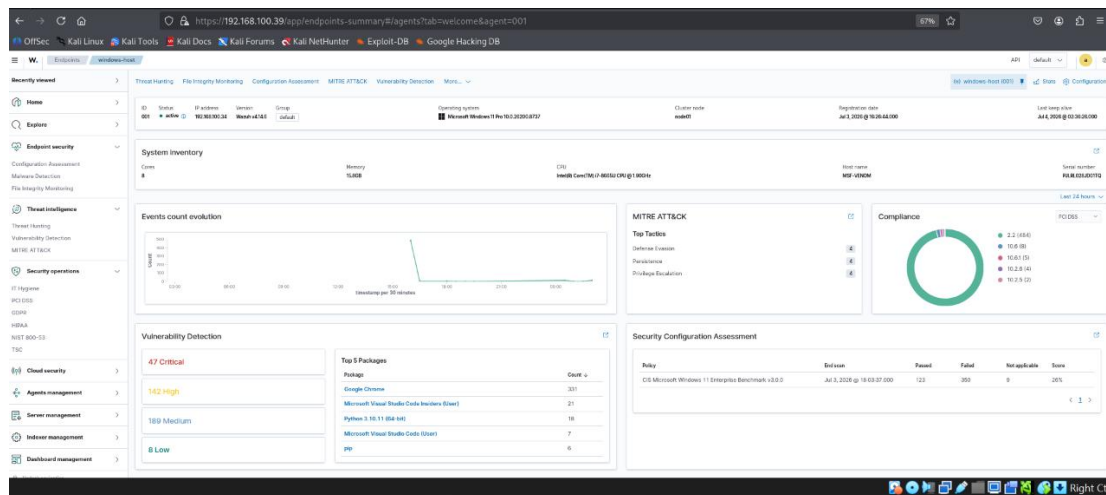


Figure 5.6: Wazuh Agent service running on the Windows endpoint.

5.5 Installation of the Wazuh Agent on Linux (Parrot Security OS)

Objective

The objective of this step was to install the Wazuh Agent on the Linux endpoint so that system logs, security events, and file integrity information could be forwarded to the Wazuh Manager for centralized monitoring.

Procedure

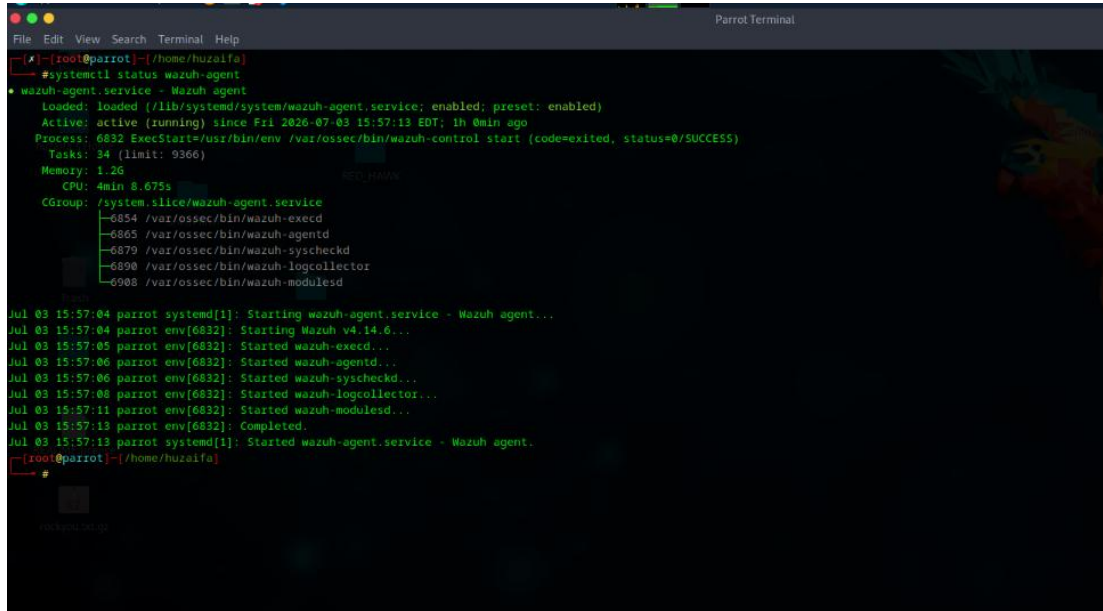
Initially, Ubuntu Server was planned as the Linux endpoint. However, due to installation issues and repeated boot failures in the virtual environment, Parrot Security OS 6.4 was selected as the Linux endpoint. Since Parrot Security OS is based on Debian, it fully supports the Wazuh Agent.

The Wazuh repository and GPG key were added to the system, after which the package list was updated. The Wazuh Agent package was installed using the APT package manager. During installation, the agent was configured with the IP address of the Wazuh Manager and assigned the name parrot-os.

After the installation, the Wazuh Agent service was enabled and started using the systemd service manager. The installation was verified by confirming that the wazuh-agent service was running successfully.

Result

The Wazuh Agent was successfully installed on the Parrot Security OS endpoint and was ready to communicate with the Wazuh Manager.



```
File Edit View Search Terminal Help
[~]root@parrot: ~/home/huzailfa
# systemctl status wazuh-agent
• wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-03 15:57:13 EDT; 1h 0min ago
     Process: 6832 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 34 (limit: 9366)
   Memory: 1.26
      CPU: 4min 8.675s
   CGroup: /system.slice/wazuh-agent.service
           └─6854 /var/ossec/bin/wazuh-execd
             └─6865 /var/ossec/bin/wazuh-agentd
               └─6879 /var/ossec/bin/wazuh-syscheckd
                 └─6890 /var/ossec/bin/wazuh-logcollector
                   └─6908 /var/ossec/bin/wazuh-modulesd

Jul 03 15:57:04 parrot systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Jul 03 15:57:04 parrot env[6832]: Starting Wazuh v4.14.6...
Jul 03 15:57:05 parrot env[6832]: Started wazuh-execd...
Jul 03 15:57:06 parrot env[6832]: Started wazuh-agentd...
Jul 03 15:57:06 parrot env[6832]: Started wazuh-syscheckd...
Jul 03 15:57:08 parrot env[6832]: Started wazuh-logcollector...
Jul 03 15:57:11 parrot env[6832]: Started wazuh-modulesd...
Jul 03 15:57:13 parrot env[6832]: Completed.
Jul 03 15:57:13 parrot systemd[1]: Started wazuh-agent.service - Wazuh agent.
[~]root@parrot: ~/home/huzailfa
#
```

Figure 5.7: Successful installation and execution of the Wazuh Agent on Parrot Security OS.

5.6 Registration of the Linux Agent

Objective

The purpose of this step was to register the Linux endpoint with the Wazuh Manager so that it could securely send monitoring data and receive management policies.

Procedure

After the Wazuh Agent service was started, the Linux endpoint automatically established communication with the Wazuh Manager using the configured manager IP address. The Wazuh Dashboard was accessed through the web interface, and the Endpoints section was opened to verify the registration status. The registered Linux endpoint appeared with the following information:

Property	Value
Agent Name	parrot-os
Operating System	Parrot Security OS 6.4
Status	Active
Manager	node01

The successful registration confirmed that the Linux endpoint was actively communicating with the Wazuh Manager.

Result

The Linux endpoint was successfully registered and appeared as Active in the Wazuh Dashboard alongside the Windows endpoint.

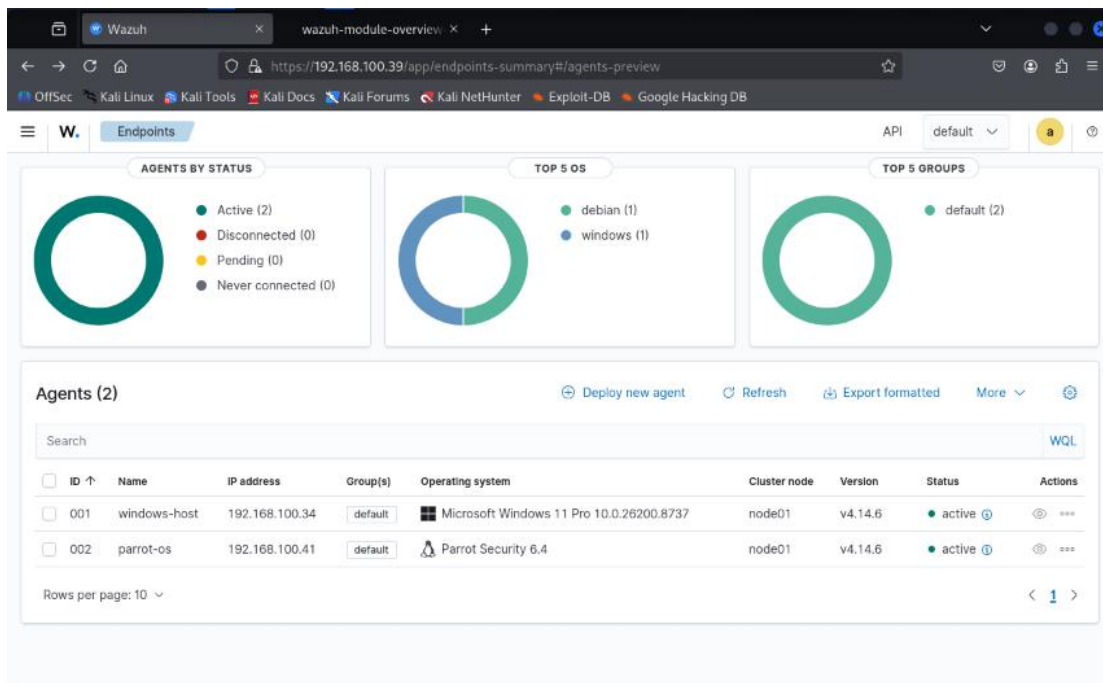


Figure 5.8: Windows and Linux endpoints successfully registered with the Wazuh Manager.

5.7 Configuration of File Integrity Monitoring (FIM)

Objective

File Integrity Monitoring (FIM) was configured to detect unauthorized file creation, modification, and deletion within a monitored directory. FIM is an important security feature that enables SOC analysts to identify unexpected changes that may indicate malicious activity or unauthorized access.

Procedure

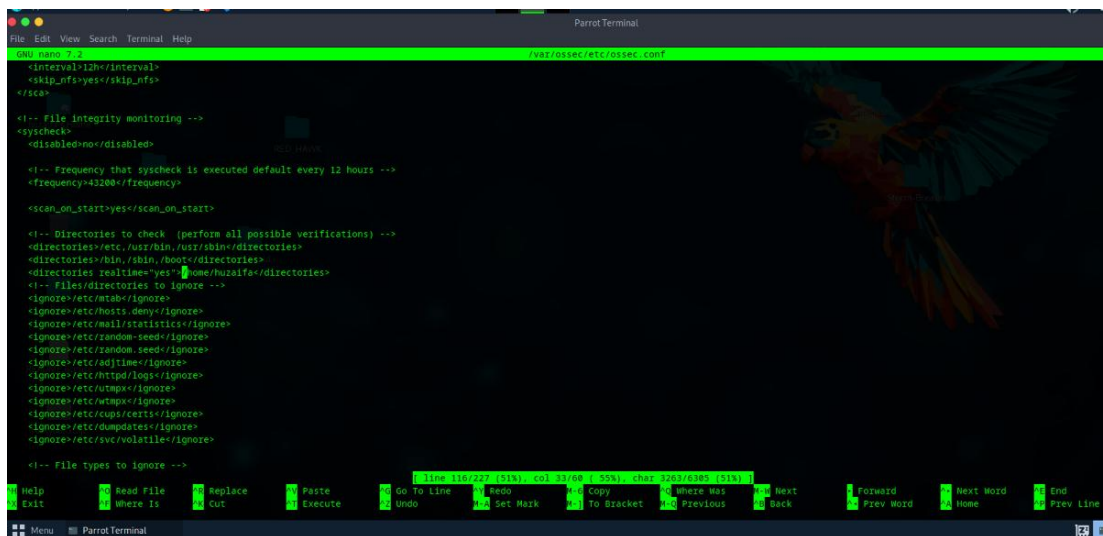
The Wazuh Agent configuration file (ossec.conf) on the Linux endpoint was modified to enable real-time monitoring of the user's home directory. The following directory was added to the <syscheck> configuration section:

```
<directories realtime="yes">/home/huzaifa</directories>
```

After saving the configuration file, the Wazuh Agent service was restarted to apply the changes. The updated configuration instructed the agent to continuously monitor the specified directory and immediately report any file changes to the Wazuh Manager.

Result

File Integrity Monitoring was successfully configured. The Wazuh Agent began monitoring the specified directory in real time and was ready to generate alerts whenever monitored files were created, modified, or deleted.



```
GNU nano 2.9.2 /var/ossec/etc/ossec.conf

<!-- File Integrity monitoring -->
<syscheck>
  <disabled>no</disabled>
  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>
  <directories realtime="yes">/home/huzaifa</directories>
  <!-- Files/directories to ignore -->
  <ignore>/etc/passwd</ignore>
  <ignore>/etc/shadow</ignore>
  <ignore>/etc/mail/statistics</ignore>
  <ignore>/etc/random_seed</ignore>
  <ignore>/etc/adjtime</ignore>
  <ignore>/etc/httpd/logs</ignore>
  <ignore>/etc/utmp</ignore>
  <ignore>/etc/cups/certs</ignore>
  <ignore>/etc/dumpdates</ignore>
  <ignore>/etc/passwd</ignore>

  <!-- File types to ignore -->
```

Figure 5.9: File Integrity Monitoring configured for the /home/huzaifa directory.

5.8 Generation and Verification of Security Events

Objective

The purpose of this step was to verify that the configured File Integrity Monitoring (FIM) functionality was operating correctly by generating file system events and confirming that they were detected by the Wazuh Manager.

Procedure

A test file named test3.txt was created inside the monitored directory. The file was then modified by adding content and finally deleted. These actions simulated normal file system activity that should be detected by the Wazuh Agent.

The generated events were transmitted to the Wazuh Manager, analyzed using the built-in detection rules, and displayed in the Wazuh Dashboard. The Threat Hunting module was used to search for the generated events and verify that the alerts were successfully recorded. The dashboard reported the following File Integrity Monitoring events:

- File Created
- File Modified
- File Deleted

These events appeared under the syscheck event group, confirming that the File Integrity Monitoring configuration was functioning correctly.

Result

The generated file system events were successfully detected by Wazuh. The alerts appeared in the Wazuh Dashboard under the Threat Hunting section, confirming successful communication between the Linux endpoint and the Wazuh Manager.

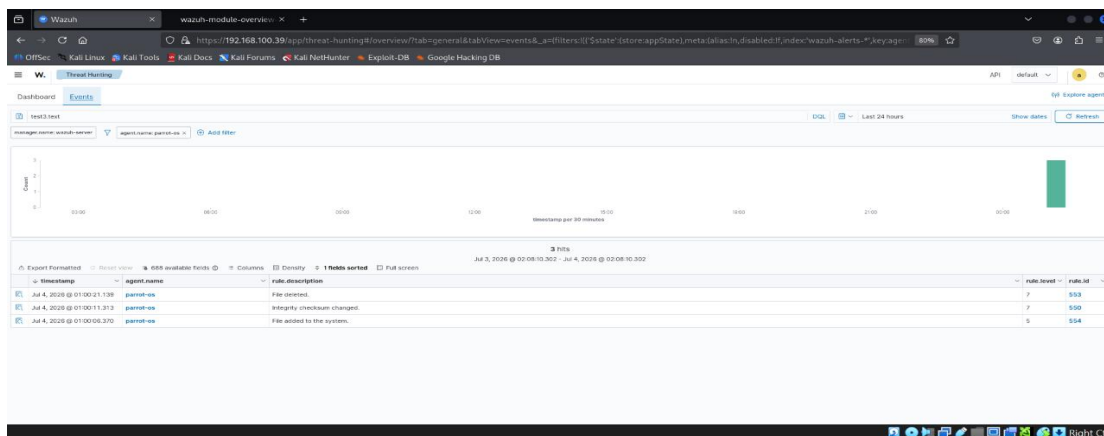


Figure 5.10: File Integrity Monitoring alerts generated after creating, modifying, and deleting test3.txt.

6. Challenges Faced

During the implementation of the Wazuh SIEM environment, several technical challenges were encountered. These challenges provided valuable troubleshooting experience and improved understanding of SIEM deployment and endpoint management.

6.1 Ubuntu Server Installation Issues

The initial plan was to use Ubuntu Server as the Linux endpoint. However, the installation repeatedly became stuck during the boot and installation process in Oracle VirtualBox. Despite updating VirtualBox and modifying virtual machine settings, the issue persisted. To continue the project within the available time, Parrot Security OS 6.4 was selected as an alternative Linux endpoint because it is Debian-based and fully compatible with the Wazuh Agent.

6.2 Windows Agent Registration

During the Windows agent installation, the agent initially did not appear in the Wazuh Dashboard. Network connectivity was verified using the ping command, and the Wazuh service was restarted. After confirming the correct manager IP address and restarting the service, the Windows endpoint successfully registered with the Wazuh Manager.

6.3 Linux Agent Configuration

The Linux agent installation completed successfully; however, File Integrity Monitoring (FIM) alerts were not generated initially. Investigation revealed that the monitored directory had not been added correctly to the <syscheck> section of the ossec.conf file. After updating the configuration and restarting the Wazuh Agent, file events were detected successfully.

6.4 Verification of File Integrity Monitoring

After configuring File Integrity Monitoring, additional testing was required to verify that file creation, modification, and deletion events were being forwarded correctly to the Wazuh Manager. The Threat Hunting module confirmed that all generated events were successfully detected. These troubleshooting activities improved practical knowledge of Wazuh configuration, Linux system administration, and SIEM deployment.

7. Learning Outcomes

The successful completion of this project provided practical experience in deploying and configuring a Security Information and Event Management (SIEM) solution using Wazuh. The following learning outcomes were achieved:

- Developed an understanding of SIEM architecture and centralized security monitoring.

- Successfully deployed the Wazuh Virtual Appliance using Oracle VirtualBox.
- Configured the Wazuh Manager as the central monitoring server.
- Installed and configured Wazuh Agents on both Windows and Linux endpoints.
- Established secure communication between endpoints and the Wazuh Manager.
- Learned how endpoint agents collect logs and forward security events.
- Configured File Integrity Monitoring (FIM) for real-time file monitoring.
- Generated simulated security events by creating, modifying, and deleting files.
- Verified that Wazuh successfully detected and displayed File Integrity Monitoring alerts.
- Gained practical experience in troubleshooting agent registration, network connectivity, and configuration issues.
- Improved understanding of SOC operations and the workflow followed by security analysts during security monitoring.

The project strengthened both theoretical knowledge and hands-on skills related to endpoint security, event monitoring, and centralized log analysis.

8. Conclusion

This project successfully demonstrated the deployment and configuration of a centralized Security Information and Event Management (SIEM) solution using Wazuh. A Wazuh Virtual Appliance was deployed as the central monitoring server, and both Windows and Parrot Security OS endpoints were successfully integrated using Wazuh Agents.

Secure communication between the endpoints and the Wazuh Manager was established, allowing the collection and analysis of security events from multiple systems. File Integrity Monitoring (FIM) was configured on the Linux endpoint to monitor changes within a selected directory.

Test events involving file creation, modification, and deletion were generated to validate the monitoring configuration. These events were successfully detected, processed by the Wazuh Manager, and displayed in the Wazuh Dashboard, confirming that the SIEM environment was functioning correctly.

Overall, this project provided valuable hands-on experience in SIEM deployment, endpoint monitoring, event analysis, and security operations. The knowledge gained through this implementation closely reflects the tasks performed by Security Operations Center (SOC) analysts in real-world enterprise environments.

9. References

The following resources were used during the implementation of this project:

- Wazuh Documentation. Wazuh Official Documentation. <https://documentation.wazuh.com/>
- Wazuh Packages Repository. Installation Packages and Repository Information. <https://packages.wazuh.com/>
- Oracle Corporation. Oracle VM VirtualBox User Manual. <https://www.virtualbox.org/>
- Microsoft Learn. Windows PowerShell Documentation. <https://learn.microsoft.com/powershell/>

- Parrot Security. Official Documentation. <https://www.parrotsec.org/>
- ITSimplera Institute. Blue Team Internship Assignment Instructions.

10. Appendix

Appendix A – Important Commands Used During the Project

Windows Agent Installation

```
Invoke-WebRequest -Uri "https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.4-1.msi" -OutFile "$env:TEMP\wazuh-agent.msi"
msiexec.exe /i "$env:TEMP\wazuh-agent.msi" /q
WAZUH_MANAGER="192.168.100.39" WAZUH_AGENT_NAME="windows-host"
NET START WazuhSvc
```

Linux Agent Installation

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --dearmor -o /usr/share/keyrings/wazuh.gpg --yes
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt stable main" > /etc/apt/sources.list.d/wazuh.list
apt update
WAZUH_MANAGER="192.168.100.39" WAZUH_AGENT_NAME="parrot-os" apt install -y wazuh-agent
systemctl enable wazuh-agent
systemctl start wazuh-agent
```

File Integrity Monitoring Test

```
touch /home/huzaifa/test3.txt
echo "Hello Wazuh" >> /home/huzaifa/test3.txt
rm /home/huzaifa/test3.txt
```